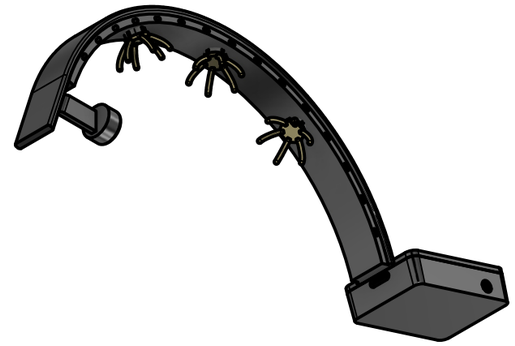
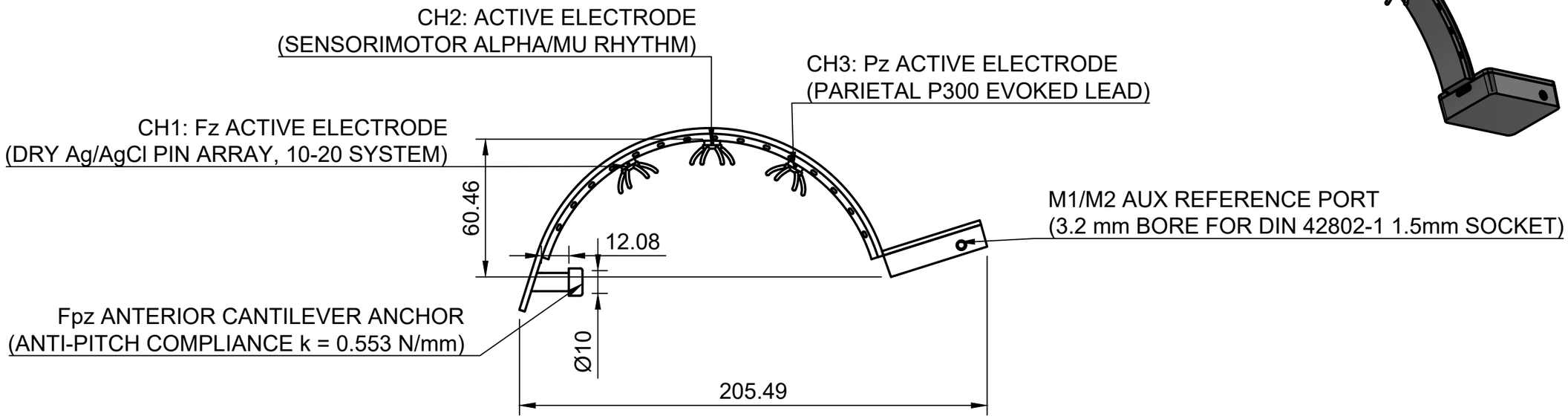
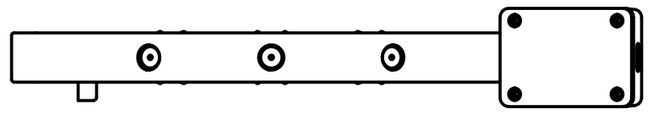








GENERAL ENGINEERING & DESIGN NOTES:

1. INTENDED DESIGN & FABRICATION CONCEPT:
- PROPOSED CHASSIS ARCHITECTURE DESIGNED FOR SLS ADDITIVE MANUFACTURING.
- INTENDED MATERIAL BASELINE: POLYAMIDE 12 (PA12 NYLON, ISOTROPIC ESTIMATE).
- SAGITTAL INTERNAL WIRE ROUTING CHANNELS MODELED FOR LEAD ISOLATION.

2. ELECTRODE INTEGRATION & ANATOMY:
- ELECTRODE MOUNT LOCATIONS ALIGNED TO 10–20 REFERENCE TOPOGRAPHY (Fz, Cz, Pz).
- HUBS DESIGNED TO HOUSE DRY MULTIPRONGED ELECTRODE HUBS (Ag/AgCl COATED).
- GEOMETRIC SCALING BASED ON STANDARD SAGITTAL CRANIAL MEASUREMENTS.

3. FINITE ELEMENT ANALYSIS (FEA) SIMULATION RESULTS:
- LINEAR STATIC STRESS SIMULATION CONDUCTED IN AUTODESK FUSION.
- BOUNDARY CONDITIONS: 1.2 N VERTICAL SAGITTAL APPOSITION LOAD APPLIED.
- SIMULATED MAX VON MISES STRESS: 11.55 MPa (VS. PA12 BULK YIELD: 48.0 MPa).
- SIMULATED FACTOR OF SAFETY: SF = 4.16; MAX DISPLACEMENT: 2.171 mm.
- SIMULATION-DERIVED STRUCTURAL COMPLIANCE: k ≈ 0.553 N/mm.
- NOTE: RESULTS ARE COMPUTATIONAL PREDICTIONS; PHYSICAL CONTACT PRESSURE AND ISOTROPY REQUIRE SUBSEQUENT BENCHMARK TESTING.

4. OCCIPITAL ENCLOSURE SPECIFICATIONS:
- PACKAGING SIZED TO HOUSE COMPACT BIOPOTENTIAL AFE, BLE SOC, AND LiPo CELL.
- BILATERAL PORTS (Ø3.2 mm BORE) SIZED FOR STANDARD 1.5 mm TOUCH-PROOF RECEPTACLES (DIN 42802-1 FORM FACTOR).

5. GENERAL TOLERANCES & SCOPE:
- PROTOTYPE CAD CONCEPT MODEL; UNTOLERANCED DIMENSIONS: ISO 2768-m.
- FABRICATION, IMPEDANCE TESTING, AND CLINICAL VALIDATION PENDING.

Dept.	Technical reference	Created by	Approved by
		Prerna Salya 10-10-2026	
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